

IAL Chemistry Unit 4 (WCH14/01) — Concise A* Study Note

Rates, Equilibria and Further Organic Chemistry · built from 21 past papers and mark schemes, June 2019 → June 2026

The examiners' **ACCEPT / IGNORE / REJECT** rulings are kept wherever they decide a mark. June and October 2019 are the legacy WCH04 papers, used where the content is the same.

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0. THE PAPER AT A GLANCE

Time / marks	1 h 45 min, 90 marks (~1.1 min per mark)
Section A	20 MCQ marks — aim for ≤ 20 minutes
Section B	~48–52 marks of structured questions, including one 6-mark levels-marked question
Section C	~18–22 marks — one long data / context question (equilibria, acids, entropy, Born–Haber)

On virtually every paper: an **entropy / feasibility calculation** · a **Born–Haber or solution cycle** · an **initial-rates table** → **rate equation** → **k with units** · an **Arrhenius or half-life graph** · **K_p or K_c** from moles · **weak-acid pH** and **buffer pH** · the **HCN + carbonyl mechanism** with the **racemic** explanation · **carbonyl tests** · **NMR deduction**.

The 6-mark levels-marked question

Indicative points seen	Marks	+ Reasoning (coherent, linked)
6	4	2
5–4	3	1 if partly linked
3–2	2	0 if unstructured
1	1	incorrect chemistry loses reasoning marks

Write **six separate points**, each a full sentence, in a logical order. All the 6-mark topics seen are listed in §33.

PART A — KINETICS

1. Orders and the rate equation from an initial-rates table

Method (3–4 marks):

1. Pick two experiments where **only one concentration changes**. Concentration $\times 2 \rightarrow$ rate $\times 1$ (order 0), $\times 2$ (order 1), $\times 4$ (order 2).
2. When **two concentrations change at once**, divide out the effect of the one you already know.
Example: $[\text{BrO}_3^-] \times 3$ alone would give rate $\times 3$; the rate went $\times 12$, so doubling $[\text{H}^+]$ gave $\times 4 \rightarrow$ **second order in H^+** .
3. **Quote the experiment numbers** and the factors — "orders without justification" scores 1.
4. Write **rate = $k[\text{A}]^m[\text{B}]^n$** — **square brackets** (round brackets penalised); include **k**.

k and its units — rearrange, substitute one experiment, then work out units:

Overall order	Units of k
0	$\text{mol dm}^{-3} \text{s}^{-1}$
1	s^{-1}
2	$\text{dm}^3 \text{mol}^{-1} \text{s}^{-1}$
3	$\text{dm}^6 \text{mol}^{-2} \text{s}^{-1}$
4	$\text{dm}^9 \text{mol}^{-3} \text{s}^{-1}$

Units usually earn **their own mark**. Answers at **1 s.f.** are penalised.

Recurring examples: rate = $k[\text{BrO}_3^-][\text{Br}^-][\text{H}^+]^2$ (k in **$\text{dm}^9 \text{mol}^{-3} \text{s}^{-1}$** , three papers) · rate = $k[\text{NO}]^2[\text{O}_2]$ or $k[\text{NO}]^2[\text{H}_2]$ (**$\text{dm}^6 \text{mol}^{-2} \text{s}^{-1}$**) · rate = $k[\text{H}_2\text{O}_2][\text{I}^-]$ · propanone iodination rate = $k[\text{CH}_3\text{COCH}_3][\text{H}^+]$ (**zero order in I_2**).

2. Graphs and half-life

Graph	Zero order	First order	Second order
concentration vs time	straight line down	curve with constant half-life	curve, half-life increases

Graph	Zero order	First order	Second order
rate vs concentration	horizontal line	straight line through origin	upward curve (straight vs $[A]^2$)

- **Initial rate:** plot concentration vs time, **draw a tangent at $t = 0$** , find its **gradient**. ("Gradient of the graph" alone is rejected.)
- **Half-life:** show **two successive half-lives on the graph** (working on the graph is required); equal $\rightarrow$ **first order**. Beware curves that level off above zero — halve towards the **final** value only if asked.
- **First order:** $k = \ln 2 \div t_{1/2} = 0.693 \div t_{1/2}$ — convert minutes to **seconds** (k in min^{-1} is not accepted as s^{-1}).
- "Constant half-life" is evidence from a **concentration–time** graph; "straight line through origin" is evidence from a **rate–concentration** graph — don't swap them.
- **Excess reagent:** "its concentration stays effectively constant, so the rate depends only on [other reactant]".

3. Following a reaction — practical methods

Method	Use when
colorimetry	a coloured species forms or is used up (Br_2 , I_2 , NO_2 brown)
gas volume / mass loss	a gas is given off
pressure / volume change	gas moles change (not for $\text{H}_2 + \text{I}_2 \rightarrow 2\text{HI}$ — no change)
conductivity	ions are formed or used
quench + titrate	samples removed; not continuous. Quench with NaHCO_3 (removes H^+ catalyst), not NaOH (alkali also catalyses / reacts with I_2)
clock reaction	time for a colour to appear; rate $\propto 1/t$

Iodine clock: a small, fixed amount of **thiosulfate** reacts with I_2 as it forms ($2\text{S}_2\text{O}_3^{2-} + \text{I}_2 \rightarrow \text{S}_4\text{O}_6^{2-} + 2\text{I}^-$); when it is used up, $\text{I}_2 + \text{starch} \rightarrow \text{blue-black}$. Thiosulfate is kept small so it is used up **before other concentrations change much** $\rightarrow 1/t$ measures the **initial** rate. Colour change: **colourless** $\rightarrow$ **blue-black** (not "clear").

To show an order belongs to H^+ and not Cl^- : **vary $[\text{Cl}^-]$ (add NaCl) or use a different strong acid** and compare rates.

4. Rate equations and mechanisms

- The **rate equation contains the species in, or before, the rate-determining step (RDS)** — with the numbers of each.
 - **A one-step reaction is unlikely** when the equation has 3+ particles: "**a collision of three (or more) particles at once is very unlikely**" (reject "three moles/compounds").
 - A proposed mechanism must (1) **add up to the overall equation** (intermediates cancel) and (2) have an **RDS matching the rate equation**.
 - A species **not** in the rate equation (e.g. I_2 in propanone iodination) reacts **after** the slow step.
 - An RDS that **breaks a strong bond (C–H)** has a high E_a .
 - **Catalyst in the rate equation:** H^+ in propanone iodination; homogeneous catalysts are **in the same phase and regenerated**.
 - **Heterogeneous catalysis:** reactants **adsorb** (not "absorb") → bonds **weakened** → products **desorb**.
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5. SN1 and SN2 — rate equations and stereochemistry

	SN2 (primary)	SN1 (tertiary)
Rate equation	$\text{rate} = k[RX][OH^-]$	$\text{rate} = k[RX]$
Steps	one step via a transition state $[HO \cdots C \cdots X]^-$	slow $RX \rightarrow R^+ + X^-$, then fast $R^+ + OH^-$
Why	little steric hindrance	stable tertiary carbocation
Chiral substrate	inversion — attack from the side opposite the leaving group → product optically active	planar carbocation , attacked equally from either side → racemic mixture
Energy profile	one hump	two humps (trough = intermediate); first E_a larger = RDS

Mechanism mark points:

- **SN2:** $C^\delta+ - Br^\delta-$ dipole · arrow from the **lone pair on O of OH^-** to C · arrow from C–Br bond to Br · **transition state with partial bonds and a – charge** · products.
 - **SN1:** dipole + arrow C–Br → Br · **carbocation + Br^-** · arrow from OH^- lone pair to C^+ .
 - A mechanism must match **your own** rate equation (a mismatch caps the marks).
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6. Temperature, Arrhenius and activation energy

Why rate rises with T (3 marks): k increases · average **kinetic energy** increases · a **greater proportion of collisions have $E \geq E_a$** . (*"IGNORE more successful collisions / more energy / more frequent collisions"*.) E_a itself is **unchanged**.

Arrhenius: $\ln k = -E_a/(RT) + \ln A$

- Plot **$\ln k$ (y, no units) vs $1/T$ (x, K^{-1})**; straight best-fit line; use **$\geq$ half the grid**; points $\pm \frac{1}{2}$ square.
 - **Gradient = $-E_a/R$** (units K , negative) $\rightarrow E_a = -\text{gradient} \times R \rightarrow$ give in **kJ mol^{-1} with a + sign**.
 - Two-temperature form: **$\ln(k_2/k_1) = (E_a/R)(1/T_1 - 1/T_2)$** .
 - Answers seen: +272, +109, +114, +58.2, +355, +242, +220 (2 s.f.), +56.9, +33.2 kJ mol^{-1} . A $\pm$ range is always allowed — a **negative E_a or positive gradient loses the mark**.
 - **Kinetic stability:** feasible ($\Delta S_{\text{total}} > 0$) but **high activation energy** $\rightarrow$ too slow at room temperature (e.g. SiO_2 bonds strong; Al protected by oxide; $\text{H}_2 + \text{O}_2$ without Pt).
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PART B — ENERGETICS: LATTICE ENERGY AND ENTROPY

7. Born–Haber cycles

Definitions to know

- **Lattice energy (LE):** enthalpy change when **1 mol of ionic solid forms from its gaseous ions** (exothermic).
- **Enthalpy of atomisation:** forms **1 mol of gaseous atoms** — $\frac{1}{2}\text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$; $\frac{1}{2}\text{Br}_2(\text{l}) \rightarrow \text{Br}(\text{g})$; $\frac{1}{2}\text{I}_2(\text{s}) \rightarrow \text{I}(\text{g})$.
- **Electron affinity:** 1 mol of gaseous atoms each gain an electron: $\text{S}(\text{g}) + \text{e}^- \rightarrow \text{S}^-(\text{g})$. The **second EA is endothermic** — an electron is added to a **negative ion**, so there is **repulsion**.
- **Hydration enthalpy:** enthalpy change when **1 mol of gaseous ions is completely hydrated / forms an infinitely dilute solution** (always exothermic; reject "energy required").

Drawing the cycle (3–4 marks)

- Upward arrows: atomisation, IE_1 , IE_2 . Downward arrows: EA_1 , LE, $\Delta_f\text{H}$ (usually).
- Every level has **species with state symbols and electrons**: e.g. $\text{Ca}^{2+}(\text{g}) + 2\text{Cl}(\text{g}) + 2\text{e}^-$.
- Penalised once: missing state symbols; missing electrons; writing the **change** on the line.

Calculating (2–3 marks) — $\Delta_f\text{H} = \Delta_{\text{at}}\text{H}(\text{M}) + \text{IEs} + n \times \Delta_{\text{at}}\text{H}(\text{X}) + n \times \text{EA} + \text{LE}$.

The classic error is **forgetting $\times 2$** for 2Cl atoms / 2 EAs — it is the only TE allowed on some schemes. Answers seen: $\Delta_f\text{H}(\text{CaI}_2) -538$; $\text{IE}_2(\text{Ca}) +1148$; $\text{EA}(\text{Cl}) -348$; $\Delta_f\text{H}(\text{CaCl}_2) -799$; $\text{IE}_2(\text{Fe}) +1567$; $\Delta_f\text{H}(\text{MgO}) -552$; $\text{LE}(\text{AlCl}_3) -5471$; $\text{IE}_3(\text{Al}) +2740 \text{ kJ mol}^{-1}$.

Why MgCl_3 doesn't form: IE_3 of Mg is huge (electron from an **inner shell**), not repaid by the extra lattice energy.

8. Lattice energy trends and polarisation

Size of LE (3 marks): LE is **more exothermic** when ions are **smaller** (ionic radius — not atomic) and **more highly charged** → **stronger electrostatic attraction**. Down Group 2: M^{2+} **ionic radius increases**, same charge → **weaker attraction** → LE **less exothermic**.

Theoretical vs experimental (Born–Haber) LE

- The **theoretical** value assumes **100% ionic**, **perfect spheres / point charges**, ions in **contact**.
- If the Born–Haber value is **more exothermic** than theoretical → **covalent character**.

- Cause: a **small, highly charged cation** (high charge density) **polarises / distorts the electron cloud** of a **large anion** (easily polarised).
- Values close → **almost 100% ionic** (e.g. NaCl, NaF, MgO — **O²⁻ small, hard to polarise**).
- Big difference → covalent character (MgI₂, MgCl₂, AgBr, MgS — **S²⁻ large, easily polarised**).

Wording traps: write **ions** not atoms ("iodide" not "iodine") · never "CaI₂ is covalent" — it has **covalent character** · never intermolecular forces · use **comparatives**.

9. Enthalpy of solution and hydration

$$\Delta_{\text{solH}} = \Sigma \Delta_{\text{hydH}} - \text{LE} \text{ (i.e. } -\text{LE} + \text{hydration enthalpies of all the ions)}$$

- **Cycle:** MX(s) → M⁺(aq) + X⁻(aq), via M⁺(g) + X⁻(g). LE arrow **down** to the solid; hydration arrows **down** to aqueous ions; an endothermic Δ_{solH} arrow goes **up** from the solid.
 - Multiply by the **number of each ion** (2Cl⁻); to get **one** Δ_{hydH} , divide by the number of ions.
 - Answers seen: NH₄Cl +20; CaCl₂ -120 / -77; MgCl₂ -122; CuCl₂ -45; $\Delta_{\text{hydH}}(\text{Cl}^-)$ -364 / -362.5 / -367 kJ mol⁻¹.
 - **Hydration more exothermic** for ions with **higher charge / smaller radius (charge density)** → **stronger ion-dipole attraction** to water (not "intermolecular forces").
 - The hydrated Cl⁻ diagram: **H⁺δ⁺** of water pointing at Cl⁻; for cations, **O⁻δ⁻** points in.
 - **Calorimetry:** $q = m c \Delta T$ (mass of **water** only); $\Delta_{\text{solH}} = -q \div n$, with sign and kJ mol⁻¹.
 - **Endothermic yet it dissolves** (KCl, NH₄NO₃, NaCl): **ΔS_{system} is positive** enough to make **ΔS_{total} positive**.
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10. Entropy

Entropy = a measure of **disorder / the number of ways of distributing energy (quanta)** among particles.

- S increases **solid < liquid < gas**; with more **atoms / electrons / complexity** (SO₂ > O₂; FeCl₃ largest); with **temperature**, with **jumps at melting and (bigger) at boiling**.
- **S° = 0 only for a perfect crystal at 0 K.**
- **Predicting the sign of ΔS_{system} (2 marks):** compare **moles of gas** (or states), with **numbers**: "1 mol solid → 3 mol gas" / "gas moles fall 4 → 2". A statement with no comparison is rejected.

The calculation chain (every paper)

Step	Formula	Traps
ΔS_{system}	$\Sigma S(\text{products}) - \Sigma S(\text{reactants})$	multiply by coefficients ($\times 3$, $\times \frac{1}{2}$); $\text{J K}^{-1} \text{ mol}^{-1}$
$\Delta S_{\text{surroundings}}$	$-\Delta H \div T$	ΔH in J ($\times 1000$); T in K (+273)
ΔS_{total}	$\Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$	same units — mixing J and kJ is rejected
feasible?	$\Delta S_{\text{total}} > 0$	
minimum T	set $\Delta S_{\text{total}} = 0 \rightarrow T = \Delta H \div \Delta S_{\text{system}}$	round up (661.5 K $\rightarrow$ 662 K; 661 rejected)

- **Signs and units** carry marks; wrong units are penalised once per question. Write **$\text{J K}^{-1} \text{ mol}^{-1}$** .
- Endothermic $\Delta H \rightarrow \Delta S_{\text{surr}}$ **negative**; exothermic $\rightarrow$ **positive**.
- **Gibbs**: $\Delta G = \Delta H - T\Delta S_{\text{system}}$; **$\Delta G = -T\Delta S_{\text{total}}$** ; feasible if **$\Delta G < 0$** .

11. Entropy, feasibility and K

$$\Delta S_{\text{total}} = R \ln K \quad \Delta G = -RT \ln K$$

- $K = e^{(\Delta S_{\text{total}} / R)}$ — e.g. $\Delta S_{\text{total}} = +555 \rightarrow K \approx 10^{29}$; $\Delta S_{\text{total}} = -85.0 \rightarrow K = 3.61 \times 10^{-5}$. No TE on a wrong expression.
- ΔS_{total} **positive** $\rightarrow K > 1$; negative $\rightarrow K < 1$.
- **ΔS_{total} vs $1/T$ graph**: gradient = **$-\Delta H$** ; where the line crosses $\Delta S_{\text{total}} = 0$ is the temperature at which feasibility changes.

Effect of raising T (3 marks — Le Chatelier alone is ignored):

1. ΔS_{system} (and ΔH) are **roughly constant** with T.
2. $\Delta S_{\text{surr}} = -\Delta H/T$ becomes **less positive** (exothermic) as T rises.
3. So **ΔS_{total} decreases** $\rightarrow$ since $\Delta S_{\text{total}} = R \ln K$, **K decreases**.

"Always feasible" (3 marks): ΔH negative $\rightarrow \Delta S_{\text{surr}}$ **always positive**; ΔS_{system} **positive** $\rightarrow \Delta S_{\text{total}}$ **always positive**. (Or $\Delta G = \Delta H - T\Delta S$ always negative.)

All four cases (6-mark question, Jan 2026):

ΔH	ΔS_{system}	Feasible?
– (exo)	+	always

ΔH	ΔS_{system}	Feasible?
–	–	only at low T (when $-\Delta H/T > \Delta S_{\text{sys}} $)
+ (endo)	+	only at high T (when $\Delta S_{\text{sys}} > \Delta H/T$)
+	–	never

Kinetic vs thermodynamic stability: $\Delta S_{\text{total}} > 0$ but no reaction $\rightarrow$ **high activation energy** (kinetically stable).

PART C — EQUILIBRIA: K_c AND K_p

12. Expressions

- **K_c** uses **square brackets**; **K_p** uses **p(...)** (round or no brackets) — **square brackets are rejected for K_p**. Powers go **outside** the brackets.
 - **Omit solids** (and pure liquids in heterogeneous systems): $\text{C(s)} + \text{H}_2\text{O} \rightleftharpoons \text{CO} + \text{H}_2 \rightarrow \text{K}_c = [\text{CO}][\text{H}_2]/[\text{H}_2\text{O}]$; $2\text{Cu(NO}_3)_2\text{(s)} \rightleftharpoons 2\text{CuO(s)} + 4\text{NO}_2 + \text{O}_2 \rightarrow \text{K}_p = p(\text{NO}_2)^4 p(\text{O}_2)$; Ag_2CO_3 decomposition $\rightarrow \text{K}_c = [\text{CO}_2]$.
 - Fractional equations give **fractional powers**: $\frac{1}{2}\text{N}_2 + \frac{1}{2}\text{H}_2 \rightleftharpoons \text{NH}_3 \rightarrow \text{K}_p = p(\text{NH}_3) / (p(\text{N}_2)^{1/2} p(\text{H}_2)^{1/2})$. Halving the equation $\rightarrow$ **K becomes $\sqrt{\text{K}}$** ; doubling $\rightarrow \text{K}^2$.
 - **Units**: work them out from the expression — atm^{-2} , atm^{-1} , atm , Pa^{-1} , mol dm^{-3} , $\text{dm}^3 \text{mol}^{-1}$... **No units when they cancel** (esterification K_c , $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$).
 - **Dynamic equilibrium: concentrations stay constant; forward and backward rates are equal** (reject "concentrations equal").
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13. Calculations

K_c from moles

1. Initial moles $\rightarrow$ change (use the **equation ratio**) $\rightarrow$ **equilibrium moles**.
2. **Divide by the volume** (forgetting $\div V$ is the usual loss) — unless $\Delta n = 0$, when volumes cancel.
3. Substitute; give units.

Esterification (titration route): acid at equilibrium from the NaOH titre $\rightarrow$ ester = water = initial acid – remaining acid $\rightarrow$ alcohol by subtraction $\rightarrow \text{K}_c \approx 4.0$ (no units).

K_p from moles (4–5 marks)

equilibrium moles $\rightarrow$ total moles $\rightarrow$ mole fraction $x = n \div n(\text{total})$
 $\rightarrow$ partial pressure $p = x \times P(\text{total}) \rightarrow$ substitute $\rightarrow \text{K}_p$ with units

- Omitting the partial-pressure step or the mole fractions typically costs 1 mark. **Don't round early** ($0.909 \rightarrow 0.9$ penalised).
- If partial pressures of the others are given: **$p(\text{missing}) = P(\text{total}) - \Sigma \text{others}$** .
- Using **$pV = nRT$** for gas moles: **Pa, m³, K** ($\text{dm}^3 \times 10^{-3}$).
- **Q_p vs K_p**: calculate the quotient for a mixture; if $Q_p \neq K_p$ it is **not at equilibrium**.
- **$\text{K}_p = \text{K}_c(\text{RT})^{\Delta n}$** (Δn = gas moles products – reactants; T in K).

- Kp answers seen: $8.29 \times 10^{-3} \text{ atm}^{-2}$, $7.59 \times 10^{-5} \text{ atm}^{-2}$, 35.2 atm^{-1} , 0.484 atm^{-1} , 0.0461 atm^{-1} , $6.9 \times 10^{-5} \text{ atm}^{-2}$, 4.57 atm , 0.0120 atm^{-1} , 0.270 Pa^{-1} .

14. Changing the conditions

Change	Effect on K	Effect on position / yield
temperature	K changes — exothermic forward: $T \uparrow \rightarrow \text{K decreases}$	follows K
pressure	none	shifts to fewer gas moles (state the numbers, e.g. $3 \rightarrow 2$)
concentration	none	shifts to restore K
catalyst / surface area	none	none — forward and backward rates increase equally ; equilibrium reached faster

Pressure argued through Kp (3 marks — "Le Chatelier" alone is ignored):

1. **Kp is unchanged** (only T changes K).
2. Increasing P increases the **denominator more than the numerator** (more gas moles on the left), so the **quotient falls below Kp**.
3. The equilibrium **shifts right** until the quotient again equals Kp $\rightarrow$ **more product**.

Temperature and K: say "**K decreases because the forward reaction is exothermic**" — "shifts left" alone doesn't earn the K mark. Or use entropy (§11).

Large K $\rightarrow$ "equilibrium lies **far to the right**" (not "shifts right") — so extra pressure gains little yield.

Colour on compression: $\text{H}_2 + \text{I}_2 \rightleftharpoons 2\text{HI}$ turns **more purple** (higher concentration, **no shift** — equal gas moles); $2\text{NOCl} \rightleftharpoons 2\text{NO} + \text{Cl}_2$ ends **more yellow** as the position shifts **left** ($3 \rightarrow 2$ moles).

Industrial compromise: high T gives an acceptable **rate** though the equilibrium yield falls; high P raises yield but costs more; **recycle unreacted gases** to get a high overall yield; a catalyst gives no extra yield but faster attainment.

Esterification $\Delta H \approx 0$: the **same number and type of bonds** are broken and made $\rightarrow \Delta S_{\text{surr}} \approx 0 \rightarrow \text{K barely changes with T}$.

PART D — ACIDS, BASES AND BUFFERS

15. Brønsted–Lowry acids and bases

- **Acid = proton donor, base = proton acceptor.** Conjugate pairs differ by one H^+ .
 - **Label A1/B1/A2/B2:** $\text{HCOOH} + \text{CH}_3\text{CH}_2\text{COOH} \rightleftharpoons \text{HCOO}^- + \text{CH}_3\text{CH}_2\text{COOH}_2^+$ — the **stronger acid donates**; the weaker acid acts as the **base**.
 - **Strong** = fully dissociated; **weak** = **partially** dissociated (a significant proportion undissociated) — not about concentration.
 - **Electron-withdrawing groups** (Cl, OH) **stabilise the anion** / weaken O–H $\rightarrow$ stronger acid ($\text{CHCl}_2\text{COOH} > \text{CH}_2\text{ClCOOH} > \text{CH}_3\text{COOH}$).
 - **Diprotic acids:** H_2SO_4 first dissociation complete; the H^+ produced **suppresses the second** dissociation (shifts it left).
 - **Arrhenius acid:** produces H^+ in water.
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16. pH calculations

Type	Method	Watch for
strong acid	$\text{pH} = -\log[\text{H}^+]$	negative pH (1.25 M $\rightarrow$ -0.097; 2.0 M $\rightarrow$ -0.3); diprotic $\times 2$
strong base	$[\text{H}^+] = K_w/[\text{OH}^-]$	Ba(OH)₂, Ca(OH)₂, Sr(OH)₂ $\rightarrow$ $[\text{OH}^-] = 2c$; use the given K_w at other temperatures
dilution (strong)	$\times 10$ volume $\rightarrow$ pH changes by 1	water to add = new volume – old volume
weak acid	$[\text{H}^+] = \sqrt{K_a \times [\text{HA}]}$	convert $\text{g dm}^{-3} \rightarrow \text{mol dm}^{-3}$ first; pH must be between 1 and 7
K_a from pH	$K_a = [\text{H}^+]^2 / [\text{HA}]$	$\text{p}K_a = -\log K_a$
excess strong base / acid after mixing	moles excess $\div$ total volume	diprotic acids: 2 $\times$ moles H_2SO_4

K_a expression: $K_a = [\text{H}^+][\text{A}^-]/[\text{HA}]$, **written with the actual formulae**, square brackets, no $[\text{H}_2\text{O}]$.
Writing only $[\text{H}^+]^2/[\text{HA}]$ **as the expression** has been rejected — write the full form.

The two weak-acid assumptions (2 marks):

1. **Dissociation is negligible**, so $[HA]_{eq} = [HA]_{initial}$.
2. $[H^+]$ from water is negligible, so $[H^+] = [A^-]$.

When the acid is fairly strong ($CHCl_2COOH$) assumption 1 fails: **$[HA]_{eq}$ is lower than assumed** → **calculated $[H^+]$ too high** → **calculated pH too low** (6-mark question, June 2024).

Weak vs strong on dilution $\times 10$: HCl pH rises by **1**; CH_3COOH rises by only **~ 0.5** , because the weak acid **dissociates more** on dilution.

K_w and temperature: ionisation of water is **endothermic** → at higher T, **K_w increases**, **$[H^+]$ increases**, **pH < 7** — but water is still neutral ($[H^+] = [OH^-]$).

17. Buffers

Definition (2 marks): a solution that **resists changes in pH** (not "prevents", not "constant pH") when **small amounts of acid or alkali** are added.

How it works (3–4 marks):

1. **Large reservoir of HA and A^-** (weak acid and its salt).
2. Added H^+ : $A^- + H^+ \rightarrow HA$.
3. Added OH^- : $HA + OH^- \rightarrow A^- + H_2O$.
4. The **ratio $[HA]/[A^-]$ hardly changes**, so the pH hardly changes.

Use the **actual species** — a generic HA/ A^- answer is capped. An **ionic equation** is required for the reaction mark.

Buffer pH: $[H^+] = K_a \times [HA]/[A^-]$ or **$pH = pK_a + \log([A^-]/[HA])$**

- **Mixing:** use moles (volumes cancel) or dilute each to the total volume.
- **Partial neutralisation:** acid left = acid – OH^- added; salt formed = OH^- added. (Forgetting the subtraction costs 1.)
- **Half-neutralisation:** $[HA] = [A^-] \rightarrow pH = pK_a$. Used to find K_a from a titration curve.
- **Ratio inverted** is the usual error (–1). A buffer pH must be sensible (basic buffers > 7).
- **Reverse:** from pH and K_a , find $[A^-] \rightarrow$ moles $\rightarrow$ mass of salt.

Buffers in context

- **Blood ($CO_2 / H_2CO_3 / HCO_3^-$):** $CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$. Large reservoir of HCO_3^- removes added H^+ (forming $H_2CO_3 \rightarrow CO_2$, exhaled); the ratio stays nearly constant. Exercise $\rightarrow$ more $CO_2 \rightarrow$ equilibria shift right $\rightarrow$ pH falls.
- **Phosphate:** $H_2PO_4^- / HPO_4^{2-}$ ($pK_a \approx 7.2$); **ammonium:** NH_4^+ / NH_3 ($H^+ + NH_3 \rightarrow NH_4^+$).
- **Food:** sodium citrate + citric acid buffer keeps pH steady $\rightarrow$ prevents bacterial / fungal deterioration.
- **Enzymes** need a specific pH (otherwise denatured).

18. Titration curves

Titration	Start pH	Vertical section	Equivalence pH	Indicator
strong acid – strong base	~1	~3 → 11	7	any (methyl orange or phenolphthalein)
weak acid – strong base	~3	~7 → 11	> 7	phenolphthalein / phenol red / bromothymol blue / thymol blue
strong acid – weak base	~1	~3 → 7	< 7	methyl orange / methyl red / bromocresol green
weak acid – weak base	~3	none	~7	none suitable — no sharp change, indicator changes gradually

- **Sketch marks:** correct start and finish pH; vertical section in the right place and height; **equivalence at the right volume** (moles!); buffer region / S-shape.
- **Weak acid–strong base:** equivalence pH > 7 because the anion is **hydrolysed**: $A^- + H_2O \rightleftharpoons HA + OH^-$. Strong acid–weak base: $NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$.
- **Buffer region** = the flat part before the end point; **half-equivalence pH = pKa**.
- **Diprotic acid: two vertical sections** (at V and 2V); pKa₁ and pKa₂ at the two half-equivalence points.
- **Indicator choice:** its **pH range lies entirely within the vertical section**. A wrong indicator (methyl orange for a weak acid) changes **gradually over a large volume, well before the equivalence point**. At the end point the colour is the **mixture** (e.g. bromothymol blue green).
- Indicators are weak acids: $HIn \rightleftharpoons H^+ + In^-$; the two forms have different colours.

19. Common numbers to recognise

- pH of 0.100 M NaOH at 298 K = **13.0**; with pK_w 14.53 → 13.53.
- Mixing 0.01 mol NaOH + 0.02 mol HCl in 40 cm³ → [H⁺] 0.25 → **pH 0.6**.
- Buffer 1 acid : 4 salt, K_a 1.7 × 10⁻⁵ → **pH 5.4**; acid : salt 1 : 2, pK_a 4.77 → **5.07**.
- Weak-acid pH answers are typically **2.4–3.0**; a buffer answer between the pK_a ± 1.

20. The acid–base "why" answers

- **Why the second proton of a diprotic acid is harder to remove:** it is removed from a **negative ion** / the first H^+ **suppresses** the second dissociation.
 - **Why add a salt of a weak acid to a weak acid:** to make a buffer (reservoir of A^-).
 - **Why a mixture is not a buffer:** complete neutralisation — **no weak acid left** (e.g. 0.2 mol CH_3COOH + 0.1 mol $\text{Mg}(\text{OH})_2$).
 - **Why pH of a salt is not 7:** hydrolysis of the ion of the weak partner.
-

PART E — ORGANIC CHEMISTRY

21. Stereoisomerism and optical activity

- **Stereoisomers:** same structural formula, **different spatial (3D) arrangement** of atoms.
 - **Chiral centre:** a carbon with **four different atoms or groups**; the molecule has a **non-superimposable mirror image** (enantiomer). (Reject "four different molecules".)
 - **Optically active:** **rotates the plane of plane-polarised light**; enantiomers rotate it **in opposite directions** (reject "bends / deflects").
 - **Racemic mixture:** **equal amounts (50:50) of the two enantiomers** → rotations **cancel**, no net rotation (not just "two enantiomers").
 - n chiral centres → up to **2^n** stereoisomers. Count chiral centres carefully in big molecules (menthol 3, nandrolone 6).
 - **3D drawings:** a tetrahedral carbon with **wedge + dash**, and its mirror image. Molecules with a C=C carbon bearing two identical groups (e.g. two CH_3) have **no E/Z** isomerism.
 - **E/Z naming:** Z when the higher-priority groups are on the same side.
-

22. Carbonyl compounds — tests and reactions

Test	Positive result	Shows
2,4-DNPH (Brady's)	yellow / orange / red precipitate (reject brown)	C=O (aldehyde or ketone)
Tollens'	silver mirror (grey/black ppt)	aldehyde
Fehling's / Benedict's	blue solution → brick-red precipitate	aldehyde
acidified $\text{K}_2\text{Cr}_2\text{O}_7$, warm	orange → green	aldehyde (or $1^\circ/2^\circ$ alcohol)
$\text{I}_2 + \text{NaOH}$ (iodoform)	pale yellow precipitate of CHI_3 , antiseptic smell	$\text{CH}_3\text{CO}-$ (methyl ketone, ethanal) or $\text{CH}_3\text{CH}(\text{OH})-$
$\text{NaHCO}_3 / \text{Na}_2\text{CO}_3$	effervescence (CO_2)	$-\text{COOH}$
PCl_5	misty / steamy fumes (HCl)	$-\text{OH}$ (alcohol or acid)

- An observation mark **depends on the right reagent**. Reagent must be complete (I_2 **and** NaOH).

- **Iodoform equation:** $\text{CH}_3\text{COCH}_3 + 3\text{I}_2 + 4\text{NaOH} \rightarrow \text{CHI}_3 + \text{CH}_3\text{COONa} + 3\text{NaI} + 3\text{H}_2\text{O}$.
- **Identifying the derivative (3–6 marks):** react with 2,4-DNPH → **filter** (Buchner) → **recrystallise** (dissolve in the **minimum hot solvent**, cool, filter, rinse with **ice-cold solvent**, dry) → **measure the melting temperature** and **compare with data**. Never boiling temperature; the m.p. is of the **derivative**, not the carbonyl.

Reduction: LiAlH_4 in **dry ether** (reject aqueous) or NaBH_4 . Aldehyde → **primary** alcohol; ketone → **secondary** alcohol; carboxylic acid → primary alcohol (LiAlH_4 only). **C=C is not reduced**. A tertiary alcohol can **never** be made by reduction.

Oxidation: acidified $\text{K}_2\text{Cr}_2\text{O}_7$, **heat under reflux** → aldehyde → carboxylic acid (write with [O]). To stop at the aldehyde from a primary alcohol, **distil it off as it forms**. Ketones are not oxidised.

Why C=O is attacked by nucleophiles but C=C isn't: C=O is **polar** — nucleophiles are attracted to the δ^+ carbon; C=C is **non-polar** and its π electrons **repel** nucleophiles.

Intermolecular forces: aldehydes/ketones have **London + permanent dipole–dipole** but **no hydrogen bonds between molecules** (they H-bond **with water** — hence solubility). Alcohols and acids H-bond → higher boiling temperatures.

23. Nucleophilic addition of HCN — the most tested mechanism

Conditions: HCN with **KCN** (or KCN + acid, pH ~5–8). KCN / NaOH **increases** $[\text{CN}^-]$ — HCN is a weak acid (K_a tiny) so on its own $[\text{CN}^-]$ is very low; **CN^- is the nucleophile** in the slow step (rate = $k[\text{carbonyl}][\text{CN}^-]$). KCN is a **homogeneous catalyst**: same phase, CN^- **regenerated**.

Mechanism points (4 marks, marked as a list of ~7 points):

1. **Lone pair on the C of $:\text{CN}^-$** (with its **negative charge**) — not from N.
2. **Curly arrow from that lone pair to the carbonyl C $^{\delta+}$.**
3. **δ^+ / δ^- dipole on C=O.**
4. **Arrow from the C=O π bond to O.**
5. **Alkoxide intermediate** with a **full negative charge** on O (not δ^-) and a **lone pair**.
6. **Arrow from the O $^-$ lone pair to H of HCN** (or H^+ / H_2O).
7. **Arrow from the H–CN bond to C of CN;** product e.g. **$\text{CH}_3\text{CH}(\text{OH})\text{CN}$** , 2-hydroxypropanenitrile.

Arrows must **start at a lone pair or bond** (not at a charge); half-arrowheads penalised once; extra wrong arrows cancel right ones.

Name: "nucleophilic addition" — CN^- **donates an electron pair to the δ^+ carbon**; two molecules join into **one product**, nothing eliminated.

Why the product is racemic (2–3 marks):

- The **C=O group is trigonal planar** — say "**planar carbonyl group**" (reject "planar molecule / intermediate / carbocation").
- CN^- attacks **equally from above and below** (reject "from any side").
- → **equal amounts of the two enantiomers**, so **no optical activity**.
- **Never mention SN1 / SN2** here — it loses marks.
- **Symmetrical ketones** (propanone, pentan-3-one) give an **achiral** product — no chiral centre, so "racemic" is **wrong** for them.
- **Enzymes** are stereospecific → a single enantiomer (lactic acid in muscle).

Follow-on: hydrolyse the hydroxynitrile with **dilute HCl, heat under reflux**: $\text{CH}_3\text{CH}(\text{OH})\text{CN} + 2\text{H}_2\text{O} + \text{H}^+ \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{COOH} + \text{NH}_4^+$ (NH_3 instead of NH_4^+ loses a mark). Alkaline hydrolysis gives the carboxylate salt + NH_3 .

Other nucleophilic additions: LiAlH_4 supplies H^- — arrow from the **lone pair on H^-** to $\text{C}^{\delta+}$, alkoxide, then protonation. NH_2OH with propanone: arrows from the **N lone pair**.

24. Carboxylic acids

- **Solubility / boiling temperature:** acids **hydrogen bond** with each other and with water ($-\text{OH}$ and $\text{C}=\text{O}$ both H-bond water); long non-polar chains lower solubility (more London forces between acid molecules).
- H-bond diagram marks: **$\delta^+ \text{H} \dots \text{lone pair on } \delta^- \text{O}$** , bond angle **$\sim 180^\circ$** , labelled H-bond, dipoles shown.
- **With carbonates:** $\text{RCOOH} + \text{NaHCO}_3 \rightarrow \text{RCOONa} + \text{H}_2\text{O} + \text{CO}_2$ (effervescence). The salt is **ionic** (reject a covalent $\text{O}-\text{Na}$ bond).
- **With $\text{NH}_3(\text{aq})$:** gives the **ammonium salt**, not an amide.
- **Reduction:** LiAlH_4 in dry ether → primary alcohol.
- **With PCl_5** → acyl chloride + $\text{POCl}_3 + \text{HCl}$ (**misty fumes**); excess PCl_5 also converts **alcohol** — $\text{OH} \rightarrow -\text{Cl}$. Also PCl_3 or SOCl_2 .
- **Acid strength** increases with electron-withdrawing Cl / OH on the chain.

25. Acyl chlorides

React vigorously at room temperature, no catalyst, releasing HCl (misty fumes; **white smoke** with NH_3 / amines). Mechanism: **addition–elimination** — the nucleophile's **lone pair** attacks the δ^+ carbon.

Reagent	Product
water	carboxylic acid + HCl

Reagent	Product
alcohol	ester + HCl
ammonia (2NH ₃)	amide + NH ₄ Cl
primary amine (2RNH ₂)	N-substituted amide + RNH ₃ ⁺ Cl ⁻
secondary amine	N,N-disubstituted amide (tertiary amines do not react)
thiol RSH	thioester

Acyl chloride route vs acid + alcohol, for esters (4 marks — each point needs its explanation):

Advantage	Explanation
fast, at room temperature	lower energy cost
not an equilibrium — goes to completion	higher yield
no catalyst	easier purification

Disadvantage	Explanation
HCl is toxic / corrosive	fume cupboard / corrosion-resistant plant
acyl chloride must be made first, anhydrous	extra step; lower atom economy

26. Esters

- **Naming:** alkyl group from the **alcohol** first, then the acid part: CH₃COOCH₂CH₂CH₃ = **propyl ethanoate**; CH₃COOCH₂CH₂CH(CH₃)₂ = **3-methylbutyl ethanoate**.
- **Formation:** acid + alcohol, **concentrated H₂SO₄ catalyst**, heat — **reversible** (role of H₂SO₄ = **catalyst**, not just "proton donor").
- **Acid hydrolysis:** reversible → acid + alcohol.
- **Alkaline hydrolysis:** NaOH, heat under reflux → **carboxylate salt** + alcohol (**not** the acid).
Triglyceride + 3NaOH → glycerol + 3 salts (soap).
- **Transesterification:** the alcohol's alkyl group replaces the ester's (biodiesel = **methyl esters** of fatty acids).
- **Cyclic esters (lactones, lactide):** from **intramolecular** esterification of a hydroxy-acid, or two lactic acids → C₆H₈O₄ + 2H₂O.
- **IR:** ester C=O **1750–1735 cm⁻¹**; no broad O–H. **Esters don't react with NaHCO₃.**

27. Condensation polymers

- **Polyester:** diol + dicarboxylic acid (or diacyl chloride) → **ester links**, H₂O (or HCl) lost. Terylene = benzene-1,4-dicarboxylic acid + ethane-1,2-diol; PLA = poly(lactic acid) $-\text{[O-CH(CH}_3\text{)-CO]}-$.
- **Polyamide:** diamine + diacyl chloride / dicarboxylic acid → **amide links**.
- **Repeat unit:** show **one** repeat unit with **continuation bonds** out of both ends; check the O count (a common MCQ distractor adds an extra O).
- **Self-condensing monomer:** one molecule with both groups (HO(CH₂)₄COCl, lactic acid).
- **Mr of a polymer** from n monomers: $n \times (\text{monomer} - 18) + 18$.
- **Biodegradable** polyesters break down by **hydrolysis**. Excess NaOH gives the diol + the disodium salt of the acid.

28. Synthesis — reagents and conditions

Conversion	Reagents and conditions
1°/2° alcohol → aldehyde / ketone	K ₂ Cr ₂ O ₇ / H ₂ SO ₄ , distil off (aldehyde)
1° alcohol / aldehyde → carboxylic acid	K ₂ Cr ₂ O ₇ / H ₂ SO ₄ , heat under reflux (not HCl)
aldehyde / ketone → alcohol	LiAlH ₄ in dry ether (or NaBH ₄)
acid → primary alcohol	LiAlH ₄ in dry ether
acid → acyl chloride	PCl ₅ (or PCl ₃ , SOCl ₂) — anhydrous
aldehyde / ketone → hydroxynitrile	HCN + KCN
nitrile → acid	dilute HCl, heat under reflux (hydrolysis)
halogenoalkane → alcohol	NaOH(aq), heat under reflux
acid + alcohol → ester	conc. H ₂ SO ₄ , heat
alcohol → alkene	conc. H ₃ PO ₄ / H ₂ SO ₄ (dehydration)
alkene → alcohol	steam + H ₃ PO ₄ / H ₂ SO ₄

Yield / atom economy: % yield = $\text{actual} \div \text{theoretical} \times 100$; atom economy = $\text{Mr}(\text{desired product}) \div \Sigma \text{Mr}(\text{reactants}) \times 100$. Scale-ups: check units (g → kg).

PART F — SPECTROSCOPY AND CHROMATOGRAPHY

29. Formulae from data

- **Combustion analysis:** mass C = mass $\text{CO}_2 \times 12/44$; mass H = mass $\text{H}_2\text{O} \times 2/18$; **O by difference** → moles → simplest ratio. (Oxygen must be **dry**, or the water mass is wrong.) "Show that" questions need the working — **no working = 0**.
 - **% composition** → moles → ratio → empirical formula; $M^+ \div \text{empirical mass}$ → molecular formula.
 - **High-resolution MS:** accurate m/z distinguishes formulae with the same integer mass: C_3H_8 44.0632, $\text{C}_2\text{H}_4\text{O}$ 44.0265, CO_2 43.9898.
-

30. Mass spectrometry and IR

MS: high-energy electrons **bombard** molecules and **break bonds** → cations + radicals; **fragments must carry a + charge**.

m/z	Fragment
15	CH_3^+
29	C_2H_5^+ or CHO^+ (propanal, not propanone)
43	CH_3CO^+ (or C_3H_7^+)
45	COOH^+
57	$\text{C}_2\text{H}_5\text{CO}^+$
M^+	molecular ion (58 propanone/propanal; 86 pentan-3-one)
$M - 15$	loss of CH_3

IR (cm^{-1}):

Bond	Range
O–H alcohol	3750–3200 (broad)
O–H acid	3300–2500 (very broad)
C–H alkane / alkene	2962–2853 / 3095–3010

Bond	Range
C=O ester	1750–1735
C=O aldehyde / ketone / acid	~1740–1720 / 1720–1700 / 1725–1700
C=O acid anhydride	two peaks (~1820 and ~1750)
C=C	1669–1645

Ignore the fingerprint region unless comparing with a reference.

31. NMR

¹³C NMR

- **Number of peaks = number of carbon environments** — use **symmetry** (identical CH₃ groups count once).
- Shifts: **C=O 160–225 ppm** (ketone/aldehyde **190–225**; acid/ester ~170); **C–O 50–90**; **C–C 0–60**.
- Examples: propanone 2 peaks; pentan-3-one 3; propanal 3; 3-bromopentane 3; carvone 10.

¹H NMR

- **Number of peaks** = H environments; **area (integration)** = number of H in each; **splitting = n + 1** (n = H on **adjacent** carbons).
- **Singlet** = no H on adjacent C (CH₃CO–, CH₃O–, (CH₃)₃C–, O–H).
- **Common patterns**: CH₃CH₂– = **triplet (3H) + quartet (2H)**; (CH₃)₂CH– = **doublet (6H) + septet (1H)**; –CH₂CH₂– between groups = two triplets; CH with 4 neighbouring H = **quintet**.

Environment	δ / ppm
R–CH ₃ , R–CH ₂ – (H–C–C)	0.8–1.8
H–C–C=O (CH ₃ CO–)	1.6–2.9
H–C–O (ester O–CH ₂ , O–CH ₃ , CH–OH)	2.8–4.2
H–C–Br	3.7–4.2
O–H (alcohol)	variable ~1–5 (singlet)
–CHO	9–10
–COOH	10–12

Deduction write-ups (5–6 marks): for each piece of evidence write **data** → **inference**: "4 peaks → 4 H environments"; "areas 3:2:2:3"; "quartet at 4.1 → CH₂ next to CH₃ and bonded to O"; "singlet at 2.2 → CH₃CO with no adjacent H"; then the **structure**. Say "**the H on C...**", not the functional group name.

Distinguishing isomers by NMR (6-mark question, June 2025 / June 2026): state the **number of peaks**, **areas**, **splitting** and a **shift** for each compound, and one **MS** difference (e.g. propanal m/z 29; propanone m/z 43 only).

32. Chromatography

- **R_f = distance moved by spot ÷ distance moved by solvent front** (both from the **baseline**). Baseline **above** the solvent level; solvent front marked as a straight line.
 - **Separation depends on the relative attraction to the stationary phase vs solubility in the mobile phase.** Higher R_f → **weaker attraction to the stationary phase / more soluble in the mobile phase**.
 - A non-polar solvent (hexane) moves polar compounds **less**.
 - **Identify a compound:** run a **pure reference** under the same conditions — same R_f (TLC) or **same retention time** (GC / HPLC).
 - **GC:** carrier gas is **inert** (He, Ar, N₂); liquid stationary phase; faster flow → shorter retention times; **peak area ∝ amount**.
 - **HPLC:** liquid mobile phase under **high pressure**; solid stationary phase; with a non-polar stationary phase, **the most polar compound elutes first**; with an –NH₂ column, –OH compounds are retained longer (H-bonds).
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PART G — EXAM TECHNIQUE

33. The 6-mark levels questions seen — practise these

Series	Topic	Indicative points
Jan 2020	propanal vs propanone with oxidising / reducing agents / 2,4-DNPH	both give DNPH ppt; both reduced by LiAlH_4 ; 1° vs 2° alcohol; only propanal oxidised (orange $\rightarrow$ green); oxidation equation
Oct 2020	Haber process above 700 K	faster rate; ΔS_{surr} less positive; ΔS_{total} falls; K_p decreases; lower yield; energy cost
Jan 2021	b.p. of pentane, butanal, propanoic acid	London in all (similar electrons); butanal + dipole–dipole; acid + H-bonds; H-bond $>$ pd–pd $>$ London
June 2021	NaCl vs MgI_2 lattice energies	NaCl $\sim$ 100% ionic; MgI_2 covalent character; Mg^{2+} higher charge density, more polarising; I^- large, polarisable; Mg^{2+} double charge $\rightarrow$ stronger bonding
Oct 2021, Jan 2025	rate equations as evidence for SN1 / SN2	primary rate = $k[\text{RX}][\text{OH}^-]$ $\rightarrow$ SN2 , one step, transition state; tertiary rate = $k[\text{RX}]$ $\rightarrow$ SN1 , slow carbocation step, fast OH^- attack
Jan 2022	structure from five tests	PCl_5 fumes $\rightarrow$ OH; DNPH $\rightarrow$ C=O; Fehling's negative $\rightarrow$ ketone; dichromate negative $\rightarrow$ tertiary alcohol; chiral $\rightarrow$ 3-hydroxy-3-methylpentan-2-one
June 2022	ethanoyl chloride + water / ethanol / NH_3	HCl lost each time; lone-pair nucleophiles attack δ^+ C; vigorous at room temperature; products acid / ester / amide
Oct 2022	anhydride from MS, IR, NMR	Mr 130; two C=O peaks (1820, 1754); 5 ^{13}C environments; CH_3CO singlet; $(\text{CH}_3)_2\text{CH}$ doublet + septet
Jan 2023	muscle vs synthetic lactic acid	define optical activity; chiral C; single enantiomer vs racemic; planar C=O attacked both sides; enzymes stereospecific
June 2023	3-bromopentane from ^1H NMR	3 environments; CHBr quintet 3.7–4.2 ppm; CH_3 / CH_2 shifts; areas 1:4:6
Oct 2023	tests for four C_4 isomers	Na_2CO_3 fizz $\rightarrow$ acid; Tollens' $\rightarrow$ aldehyde; iodoform $\rightarrow$ methyl ketone / $\text{CH}_3\text{CH}(\text{OH})$; ester negative to all

Series	Topic	Indicative points
Jan 2024	experimental vs theoretical LE (NaF, MgF ₂ , MgCl ₂)	Mg ²⁺ charge density; stronger attraction; Mg ²⁺ polarising; Cl ⁻ polarisable; biggest gap MgCl ₂ → most covalent; NaF most ionic
June 2024	measured vs calculated pH of chloro-acids	[H ⁺] = √(K _a [HA]); [HA] _{eq} < initial as dissociation significant; calculated pH too low; bigger error for the stronger acid
Oct 2024	identifying a carbonyl by its 2,4-DNPH derivative	orange ppt; filter; recrystallise (min. hot solvent, cool); cold rinse; dry; m.p. vs data
June 2025 / June 2026	NMR (+ MS) of propanal vs propanone / propanone vs pentan-3-one	peak numbers, areas, splitting, shifts, C=O ¹³ C peak 190–225; m/z 29 vs 43
Oct 2025	functional groups from tests	NaHCO ₃ → COOH; dichromate → alcohol; iodoform → CH ₃ CH(OH)/CH ₃ CO; cyclic esters
Jan 2026	feasibility from ΔH and ΔS _{system}	the four-case table in §11

34. Examiner rulings that cost marks

- **Brackets:** K_c, K_a and rate equations → **square brackets**; K_p → **p(...)**, never square brackets; powers **outside** brackets; no [H₂O] in K_a; **HA / A⁻ must be defined** — use the real formulae.
- **Sig figs:** "ignore s.f. except 1 s.f." is the default — **1 s.f. always loses a mark**. Obey explicit demands ("to 3 s.f.", "2 s.f." for a tangent). Round **minimum temperatures up**. Don't round early.
- **Units:** k, K_c/K_p, ΔS, E_a, rate — often a separate mark. **J K⁻¹ mol⁻¹** (not "J/K mol"). pH has **no** units. K_c with cancelling units has **no** units.
- **Signs:** ΔS_{surr} = -ΔH/T; E_a and positive enthalpies with **+**; a sign slip usually costs 1.
- **No TE on a wrong expression or cycle.** TE is otherwise generous — **show working**.
- **Plausibility checks:** weak-acid pH between 1 and 7; excess-base pH > 7; buffer pH near pK_a.
- **Wrong chemistry alongside right chemistry is penalised:** "atoms" for ions; "chlorine" for chloride; "absorb" for adsorb; "planar intermediate" in HCN addition; "boiling temperature" for derivatives; "more hydrogen bonds" in a ketone; "clear" for colourless; "intermolecular forces" for ion-dipole.
- **Names:** exact IUPAC — "propan-1-ol" not "propanol"; "2-hydroxypropanenitrile" (reject hydroxo/hydroxyl/nitrile); if both name and formula are given, **both** must be right.
- **Graphs:** axes the right way round, labelled with units (ln k has none; 1/T in K⁻¹), **≥ half the grid**, points ±½ square, best-fit line (or smooth curve, not point-to-point), working **on the graph**.

35. The 20 most lost marks

1. Forgetting **×2** (two Cl atoms / EAs / hydration of 2Cl^-) in Born–Haber and Hess cycles.
 2. **kJ not converted to J** in $\Delta S_{\text{surr}} = -\Delta H/T$; $^{\circ}\text{C}$ not converted to K.
 3. Stoichiometric **coefficients** missed in ΔS_{system} .
 4. **Rounding the minimum temperature down.**
 5. **Not dividing by volume** for K_c ; skipping mole fractions or partial pressures for K_p .
 6. **Square brackets in K_p** ; round brackets in K_c / K_a / rate equations.
 7. Missing or wrong **units of k**; k in min^{-1} instead of s^{-1} .
 8. **Negative E_a** , or E_a left in J when kJ is asked.
 9. Orders stated **without quoting experiments**.
 10. "More successful collisions" instead of "**more collisions with $E \geq E_a$** ".
 11. Forgetting **$[\text{OH}^-] = 2c$** for $\text{Ba}(\text{OH})_2$ / $\text{Ca}(\text{OH})_2$ / $\text{Sr}(\text{OH})_2$; forgetting **$2 \times \text{H}_2\text{SO}_4$** .
 12. Excess base concentration not over the **total volume**.
 13. **Inverting** the buffer ratio.
 14. "Buffer **keeps pH constant**" instead of "**resists change**"; no ionic equations.
 15. Temperature effect on K explained only by "**shifts left**" with no mention of K.
 16. Pressure answer by Le Chatelier when the question asks **through K_p** .
 17. "**Planar intermediate / molecule**" instead of "**planar $\text{C}=\text{O}$** "; mentioning $\text{S}_{\text{N}}1/\text{S}_{\text{N}}2$ in HCN addition.
 18. Curly arrow from **N of CN^-** , or from a **charge** instead of a lone pair; no charge on the alkoxide.
 19. **Polarisation** answers using atoms, "covalent" instead of "covalent character", or no comparison.
 20. " LiAlH_4 " without **dry ether**; "distil" without "**as it forms**"; "boiling temperature" of the derivative.
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36. Per-paper map (main structured questions)

Series	Kinetics	Energetics	Equilibria / acids	Organic / analysis	6-mark
Jun 2019*	$\text{NO} + \text{H}_2$ orders, RDS	KIO_3 entropy, min T; NaCl vs KCl LE	propanoic acid pH, buffer; K_p CH_3OH	HCN mechanism; keto-ester NMR	—
Oct 2019*	$\text{S}_{\text{N}}2$ kinetics, tangent; $\text{S}_{\text{N}}1$ racemic	NH_4NO_3 ΔS , K	HCOOH pH, buffer	ester routes; IR; $\text{C}_5\text{H}_8\text{O}_3$	—

Series	Kinetics	Energetics	Equilibria / acids	Organic / analysis	6-mark
Jan 2020	Arrhenius (cyclopropane)	SiC ΔS_{total} , K; CaI_2 Born–Haber, polarisation	buffer by partial neutralisation	NMR; TLC; ester yield	propanal vs propanone
Oct 2020	—	CaCl_2 Born–Haber, ΔsolH	H_3PO_4 Ka; dilution; Haber Kp	$\text{C}_5\text{H}_{10}\text{O}_2$ isomers, NMR	Haber above 700 K
Jan 2021	BrO_3^- orders; Arrhenius	KBrO_3 entropy	SO_3 Kp; H_2SO_4 buffer	menthone; HCN; ester NMR	b.p. and IMFs
Jun 2021	BrO_3^- orders; catalysis	entropy of dissolving	Deacon Kp; blood buffer	HCN; lactic acid; ester NMR	NaCl vs MgI_2 LE
Oct 2021	half-life; Arrhenius	Born–Haber MCQs	ethanol Kp; HSO_4^- pH, buffer	combustion analysis; keto-acid NMR	SN1 / SN2 evidence
Jan 2022	BrO_3^- orders	Haber ΔS vs $1/T$, K	ester Kc; hexanoic acid pH; NH_4^+ buffer	HCN mechanism; test deduction	structure from tests
Jun 2022	SN1 kinetics	Fe_2O_3 ΔS , min T; FeO Born–Haber	Haber Kp; HCOOH titration	$\text{C}_5\text{H}_{10}\text{O}$ isomers; ^{13}C NMR	ethanoyl chloride reactions
Oct 2022	half-life; two-T Ea	SrI_2 MCQs; ΔG	HI Kc, $\ln K$ vs $1/T$	lactic acid; HCN; acyl chloride	anhydride spectra
Jan 2023	iodine clock; two-T Ea	butane cracking ΔS , Kp	gluconic acid pH, titration, buffer	lactide NMR; LiAlH_4 mechanism	lactic acid optical activity
Jun 2023	$\text{NO} + \text{O}_2$ orders	CaCl_2 Born–Haber; N_2O ΔS ; Arrhenius	Tris buffer	SN2 ; iodoform; DNPH	3-bromopentane NMR
Oct 2023	$\text{NO} + \text{O}_2$ k; mechanism check	Fe_2O_3 / Al_2O_3 feasibility	NO_2 Kc; pentanoic acid curve	HCN mechanism; NMR	tests for four isomers
Jan 2024	iodine clock; Arrhenius	MgCl_2 Born–Haber, ΔsolH	ester hydrolysis Kc; titration curve	HCN; PLA NMR	experimental vs theoretical LE

Series	Kinetics	Energetics	Equilibria / acids	Organic / analysis	6-mark
Jun 2024	tangent; half-life; Arrhenius	silicon ΔS	Ka / pKa; diprotic curve; dilution	C ₄ H ₆ O ₂ acids; iodoform; NH ₂ OH mechanism	calculated vs measured pH
Oct 2024	SN2 orders, inversion; SN1 profile	BaCO ₃ min T; MgCl ₂ calorimetry, Hess	H ₂ SO ₄ excess pH; H ₂ SO ₃ buffer	ester NMR; aspirin; HCN + NaOH	2,4-DNPH derivative
Jan 2025	clock graph; tangent	effervescent tablet ΔS ; ΔsolH diagram	oxalic acid Ka ₁ , Ka ₂	pentanone tests; ester NMR	SN1 / SN2 evidence
Jun 2025	H ₂ O ₂ titre half-life; mechanism	CuSO ₄ min T; CuCl ₂ Hess	NOCl Kc; ethanoic acid buffer	acid derivatives; H-bond	propanal vs propanone NMR/MS
Oct 2025	propanone iodination, RDS	CoCl ₂ ΔS , min T	SO ₃ Qp; Ag ₂ CO ₃ Kc; dilution weak vs strong	hydroxyketone NMR	functional groups from tests
Jan 2026	NO + O ₂ orders, RDS; Arrhenius	MgO Born–Haber; MgS polarisation	NH ₃ Kp; blood and phosphate buffers	odour molecules: MS, HPLC, IR	feasibility cases
Jun 2026	SN2 orders	AlCl ₃ ΔS ; hydration; LE	SO ₃ Kp in Pa	HCN + pentan-3-one; lactic acid	propanone vs pentan-3-one NMR

* legacy WCH04 papers.

37. A* requirements and routine

A* = grade A overall (≥ 480 / 600 UMS) **and** ≥ 270 / 300 across the A2 units (**WCH14 + WCH15 + WCH16**). Unit 4 is highly predictable — the same calculation types and mark points repeat every series.

Priorities: (1) entropy chain and feasibility · (2) Kp / Kc calculations · (3) weak acid, buffer and excess-base pH · (4) orders, k units, Arrhenius · (5) HCN mechanism + racemic · (6) Born–Haber,

Hess, polarisation · (7) carbonyl tests, acyl chlorides, esters · (8) NMR deduction · (9) the 6-mark topics in §33.

1. Section A in ≤ 20 minutes; mark doubtful MCQs and return.
2. Calculations: expression $\rightarrow$ substitution $\rightarrow$ answer $\rightarrow$ UNITS $\rightarrow$ s.f.
Check: J vs kJ, K vs $^{\circ}\text{C}$, $\times 2$ for diprotic / two ions, divide by V.
3. Explanations: one idea per sentence, with the key word
(reservoir / planar $\text{C}=\text{O}$ / $E \geq E_{\text{a}}$ / charge density / K_{p} unchanged).
4. Mechanisms: dipoles, lone pairs, arrows from lone pairs or bonds,
charges on ions and intermediates.
5. 6-mark: six separate indicative points in a logical order.